

# Example

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Growth of Hudson River striped bass; years 1965–1976.

$y_t$  = incremental growth rate in year  $t$

$x_{1t}$  = population density in year  $t$

$x_{2t}$  = rise in temperature in year  $t$

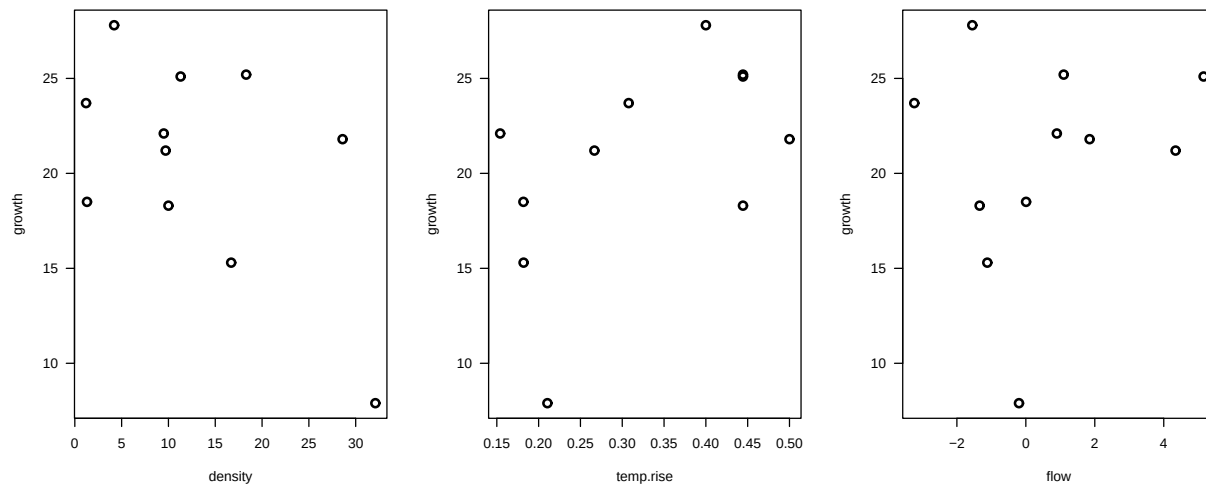
$x_{3t}$  = a measure of river flow in year  $t$

Fit the model

$$y_t = \beta_0 + \beta_1 x_{1t} + \beta_2 x_{2t} + \beta_3 x_{3t} + \epsilon_t \quad \text{where } \epsilon_t \sim \text{iid Normal}(0, \sigma^2)$$

Plot  $y$  vs each  $x$

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## Results

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|           | Est   | SE   | t-value | P value |
|-----------|-------|------|---------|---------|
| Intercept | 17.00 | 3.22 | 5.3     | 0.001   |
| density   | −0.37 | 0.11 | −3.2    | 0.015   |
| temp.rise | 25.36 | 9.01 | 2.8     | 0.026   |
| flow      | 0.45  | 0.47 | 1.0     | 0.362   |

## Diagnostics

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### Assumptions

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$\epsilon$ 's independent

$\epsilon$ 's normally distributed

$\epsilon$ 's have constant SD

y's linear in each of the x's

No other x's belong in the model

### Diagnostics

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Plot residuals vs time

QQ plot of residuals

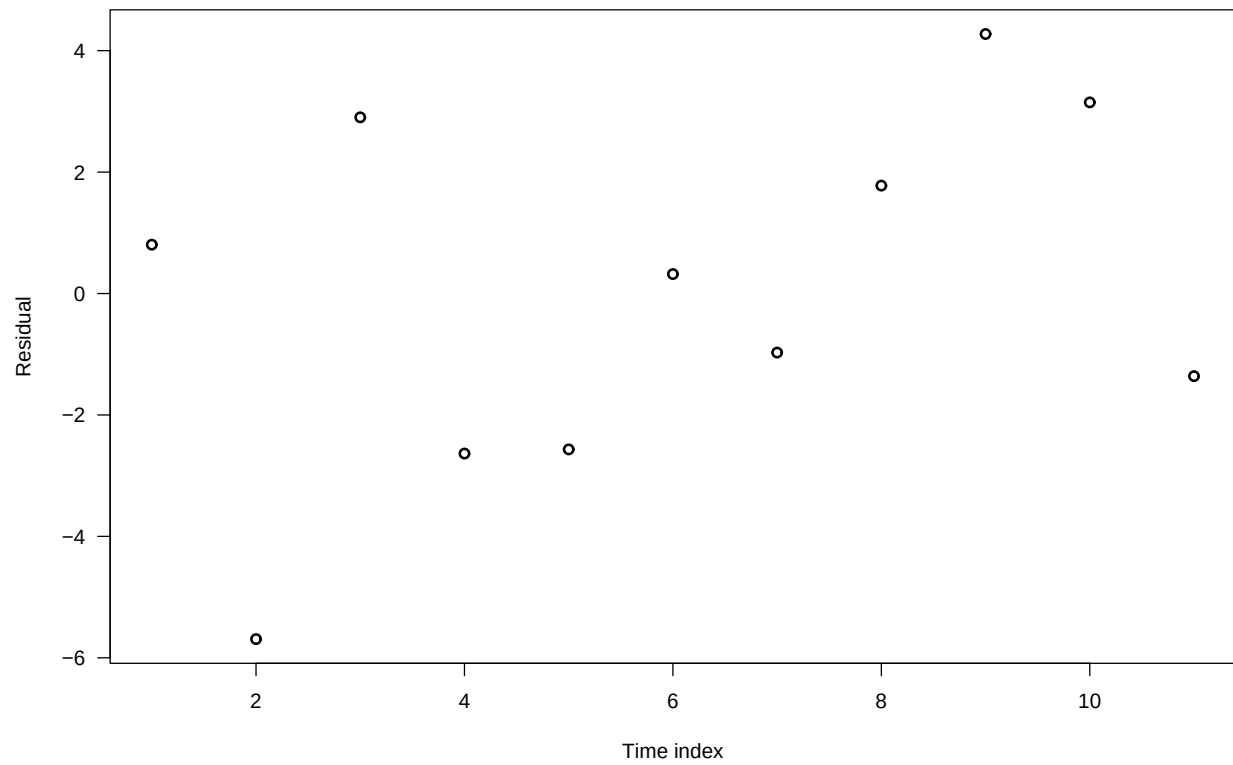
Plot residuals vs fitted values

Plot residuals vs each x

Plot residuals vs other x's

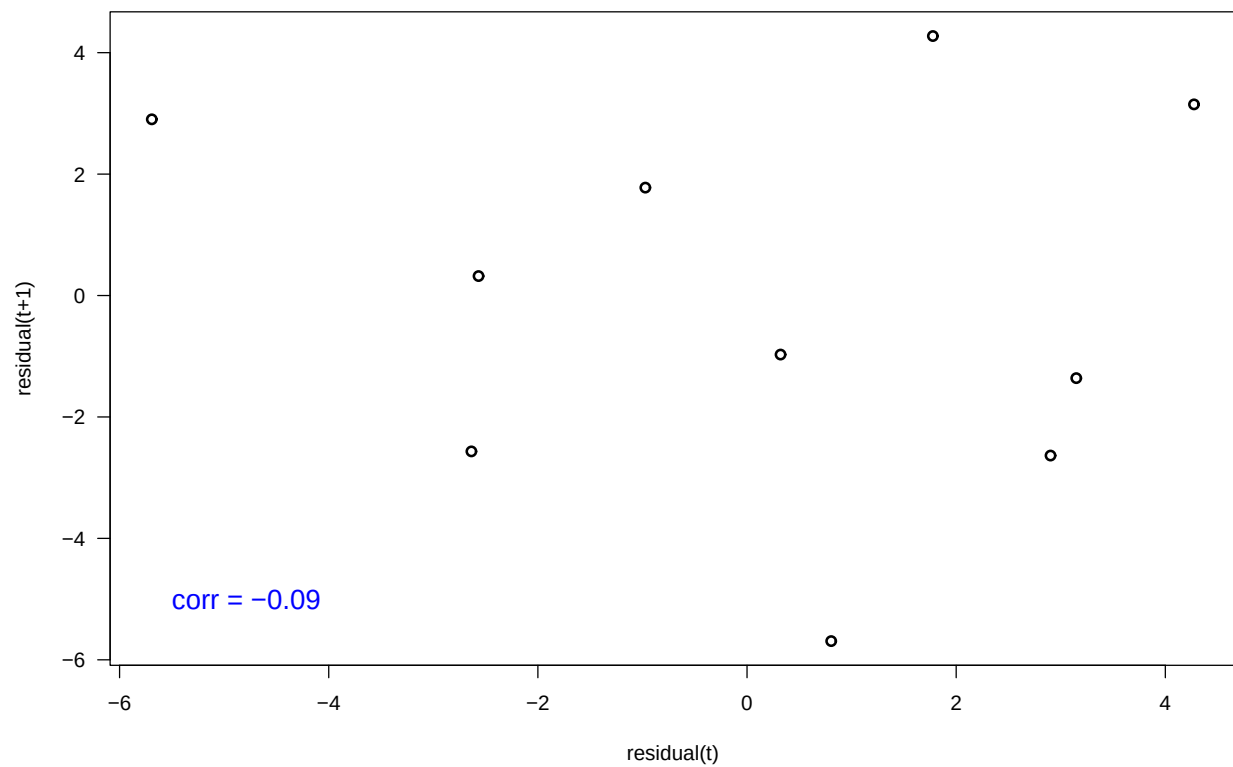
## residuals vs time

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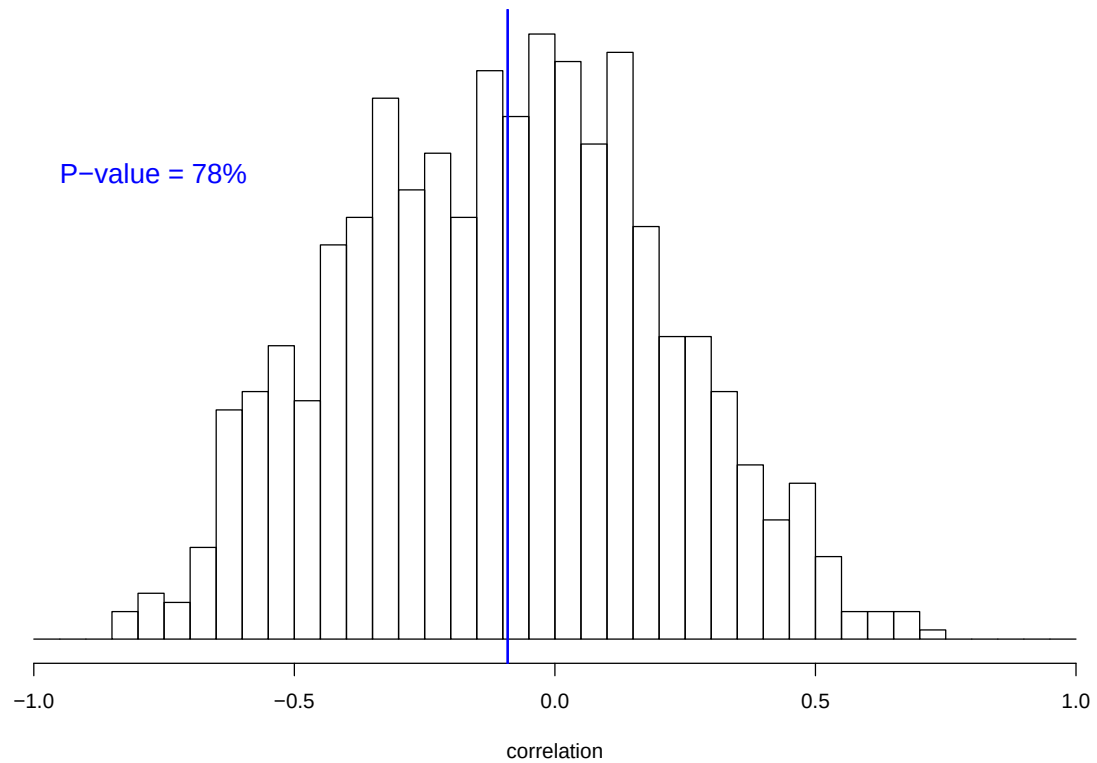
## residual(t) vs residual(t+1)

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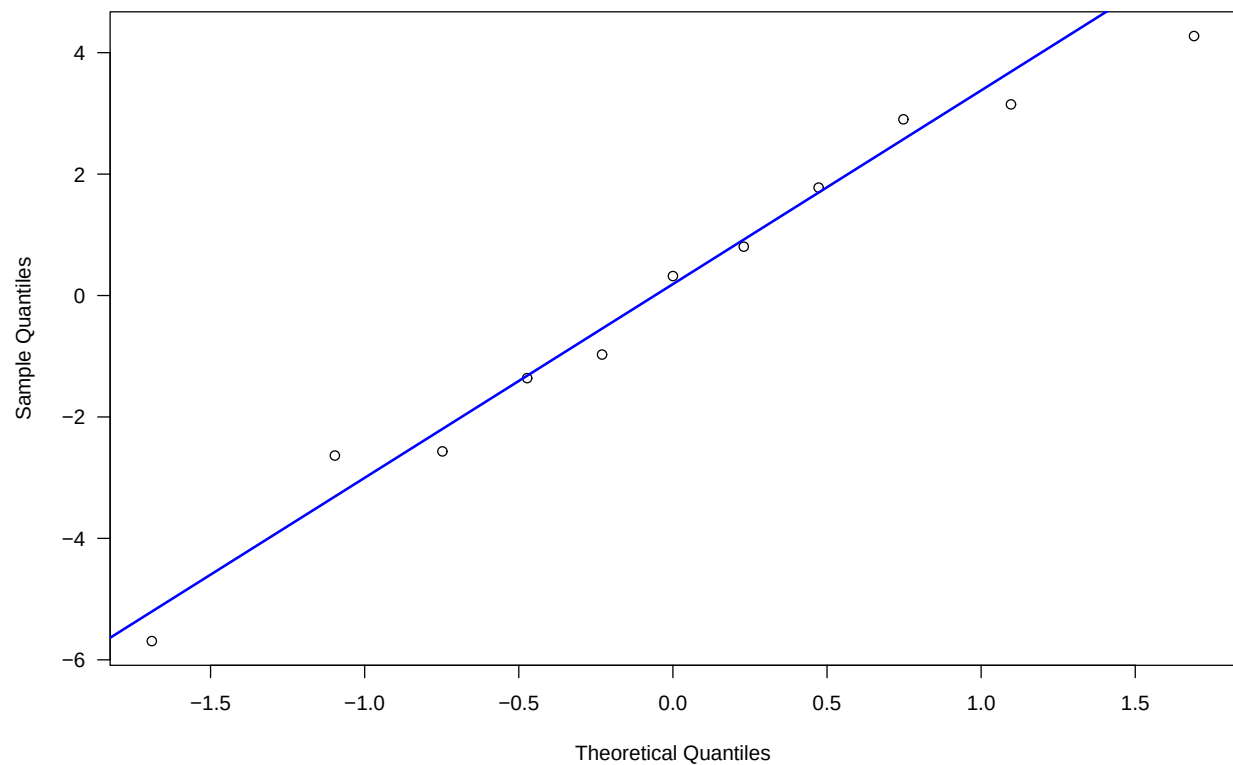
# Permutation test

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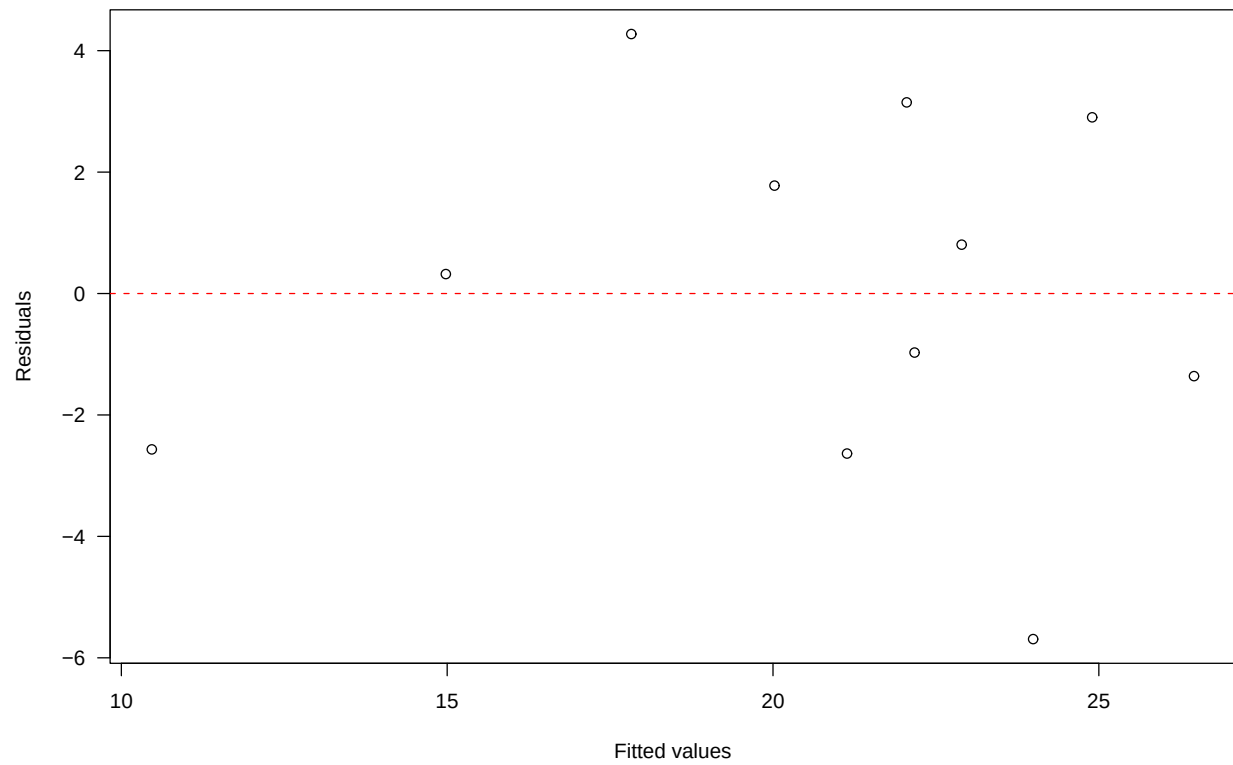
# QQ plot of residuals

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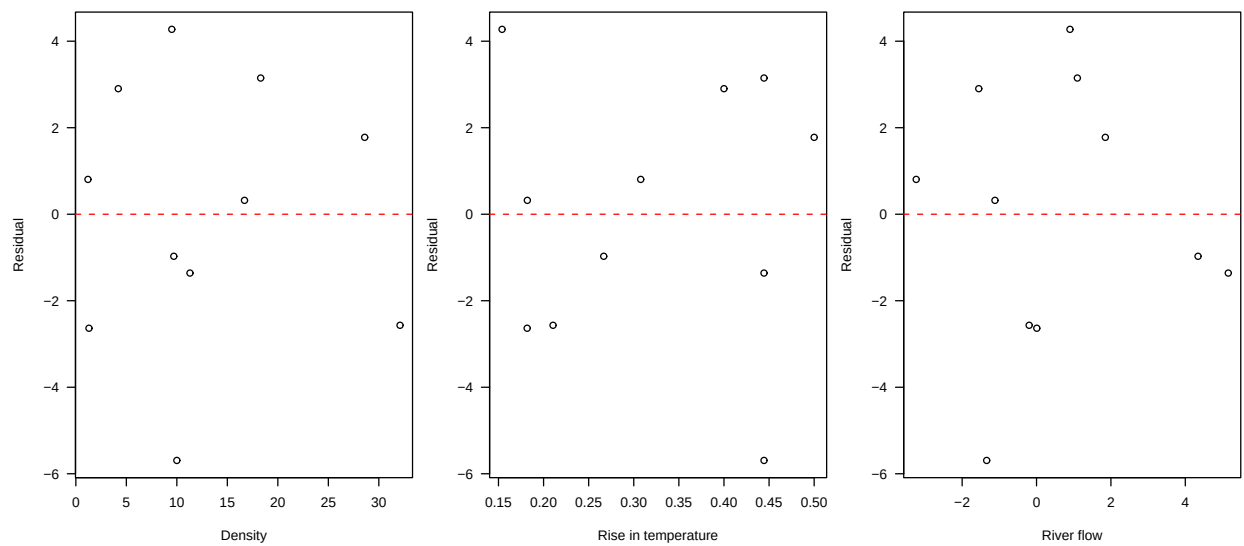
# Residuals vs fitted values

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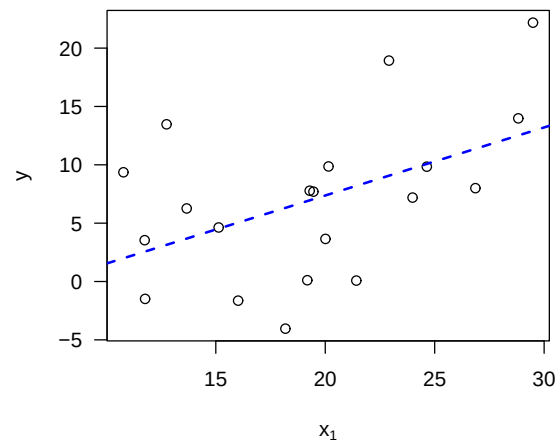
# Residuals vs x's

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# A fake example

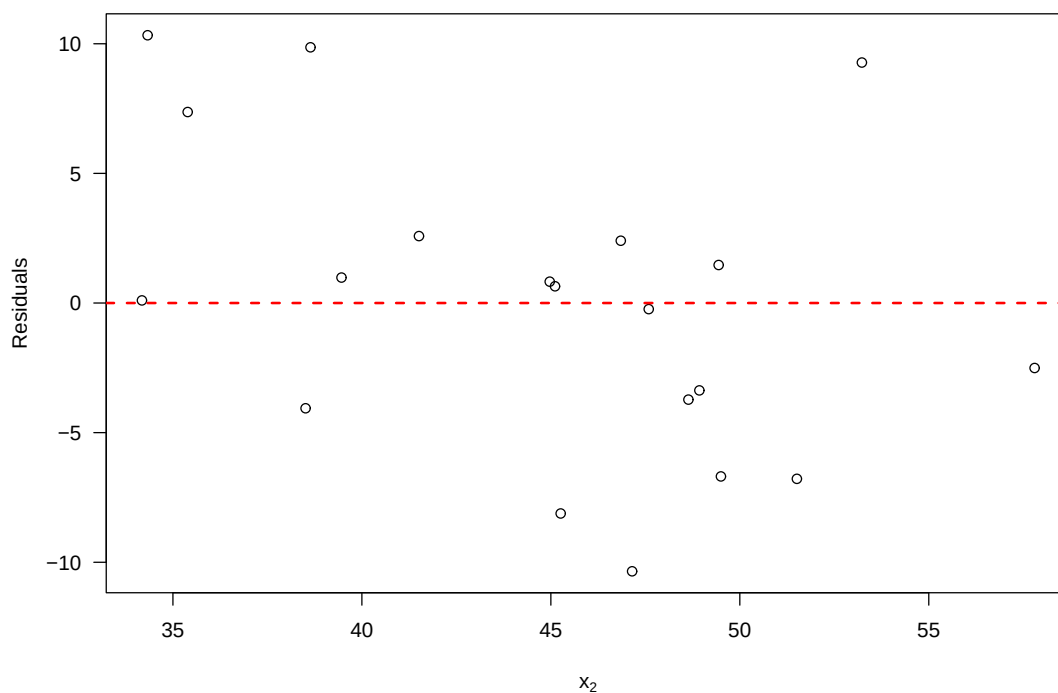
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|           | Est   | SE   | t-val | P-val |
|-----------|-------|------|-------|-------|
| Intercept | -4.29 | 4.97 | -0.86 | 0.40  |
| $x_1$     | 0.58  | 0.25 | 2.36  | 0.03  |

## Residuals vs another x

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## Regress y on both x's

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|           | Est   | SE   | t-val | P-val  |
|-----------|-------|------|-------|--------|
| Intercept | 18.02 | 8.50 | 2.12  | 0.0491 |
| x1        | 1.21  | 0.29 | 4.13  | 0.0007 |
| x2        | -0.77 | 0.26 | -3.00 | 0.0080 |

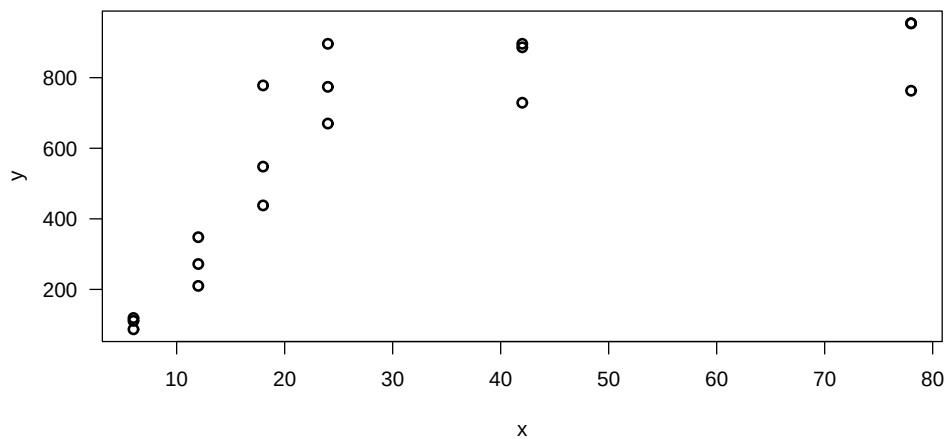
## One last example

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Sediment ingestion by the mud snail, *Hyrobia minuta*.

y = Amount ingested

x = Time allowed to eat

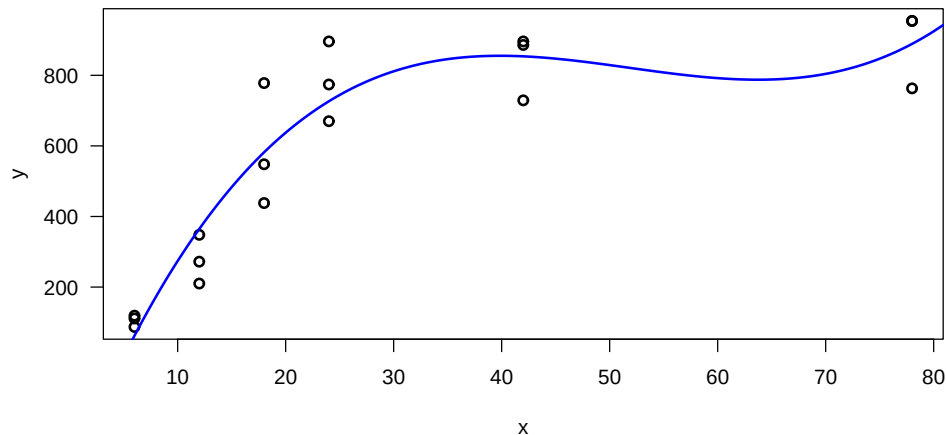


# A model

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Let's consider the model

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \beta_3 x_i^3 + \epsilon_i \quad \text{where } \epsilon_t \sim \text{iid Normal}(0, \sigma^2)$$



## Estimated coefficients

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|           | Est   | SE    | t-val | P-val  |
|-----------|-------|-------|-------|--------|
| Intercept | -339  | 127   | -2.66 | 0.019  |
| time      | 75.7  | 15.4  | 4.91  | <0.001 |
| time^2    | -1.55 | 0.48  | -3.22 | 0.006  |
| time^3    | 0.010 | 0.004 | 2.52  | 0.024  |



# Diagnostic plots

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